

Multicalibration Is Necessary for Unbiased Model-Based Prevalence Estimation

Want to say this?

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Significance

Researchers increasingly use AI systems—including large language models—as measurement devices to estimate how common a phenomenon is in a population. A critical but overlooked problem arises when the target population differs from the one on which the device was validated: standard calibration methods produce biased prevalence estimates. We show that multicalibration—requiring a device to be accurate not just on average but conditional on the input features—is necessary and sufficient for unbiased prevalence estimation under covariate shift. Our result, confirmed by simulation and empirical applications, implies that measurement in general and the rapidly growing body of research using AI for measurement specifically is vulnerable to systematic bias unless multicalibration is enforced.

Maths language, shown by a theorem and confirmed by experiments, or only confirmed by experiments?

Abstract

Estimating the prevalence of a category in a population using imperfect measurement devices—diagnostic tests, classifiers, or large language models—is fundamental to science, public health, and platform governance. Standard approaches correct for known device error rates but assume these rates remain stable across populations. We show this assumption fails under covariate shift and that multicalibration—calibration conditional on the input features, not just on average—is both necessary and sufficient for unbiased prevalence estimation under such shift, connecting recent theoretical work to a longstanding problem in measurement. A simulation confirms that standard methods exhibit bias growing with shift magnitude, while a multicalibrated estimator maintains near-zero bias. Two empirical applications—estimating employment prevalence across U.S. states using the American Community Survey, and classifying political texts across four countries using an LLM—demonstrate that multicalibration substantially reduces bias in practice, while highlighting that calibration data should cover the key feature dimensions along which target populations may differ.

Introduction

Emphasising the importance of unbiased prevalence estimation ?

Researchers in nearly every field frequently aim to measure the prevalence of a category within a target population. This includes use of diagnostic tests to measure disease prevalence, tracking prevalence of phenomena on online platforms, and quantifying document frequencies in a corpus. Recently, large language models (LLMs) have dramatically expanded the ease and reduced the cost of this kind of measurement. Researchers now routinely deploy LLMs as zero-shot measurement devices to estimate the prevalence of phenomena that previously required expensive manual annotation.

Applications span the social sciences, where LLMs code democracy indicators across countries (Weidmann, Faulborn, and Garcia 2025), classify protest events in news corpora (Overös et al. 2024), and categorize

open-ended survey responses at near-human accuracy (Mellon et al. 2024; Gilardi, Alizadeh, and Kubli 2023); clinical medicine, where they extract diagnostic attributes from pathology reports (Sushil et al. 2024) and identify goals-of-care discussions in clinical notes (Sibley et al. 2025); and the digital humanities, where they annotate art forms in auction records (Tojima and Yoshida 2025) and convert qualitative text into quantitative variables across multiple languages (Karjus et al. 2025). LLMs are also being advertised for content moderation¹ and LLM “judges” are used to assess the quality and safety of AI systems (Zheng et al. 2023; Yuan et al. 2024). The central promise of these model-based measurements is extrapolation: the ability to estimate prevalence in new populations, time periods, or subgroups where manual annotation is infeasible due to scale or too expensive using traditional human annotation. While model-based quantification is more affordable, it typically comes with the risk of false positive and false negative classification errors. However, commonly used remedies for these errors fail when the target population differs from the population on which the device’s error properties were assessed.

A variety of methods to correct imperfect devices for the purpose of measurement exist. So-called quantification methods, such as “Classify & Count”, “Adjusted Count”, or the Saerens-Latinne-Decaestecker EM algorithm, and related methods, offer adjustments that attempt to correct for imperfect devices (González et al. 2017). However, these methods rely on a fragile assumption: that the device’s error properties (sensitivity, specificity, or calibration) remain static across populations. In reality, a model’s performance often varies drastically across subgroups. When the composition of the target population differs from the source population, these subgroup-specific errors no longer cancel out, leading to biased prevalence estimates.

We propose multicalibration as the necessary and sufficient condition for unbiased prevalence estimation under covariate shift. Building on recent theoretical advances by Kim et al. (2022), we leverage the fact that a multicalibrated classifier achieves ‘Universal Adaptability’ to solve the prevalence estimation problem. We show that this theoretical guarantee allows for unbiased estimates on any downstream target population without the need for target-specific re-weighting or retraining. This robustness is critical for nearly all practical applications of quantification, including applying a model to previously unseen datasets, tracking changes in prevalence over time, or analyzing subsets of the training population. In contrast, we show that standard calibration and adjustment-oriented quantification techniques are insufficient for unbiased estimation except under restrictive assumptions.

We review existing approaches to the quantification problem, connect the quantification task to the broader theory of domain adaptation via multicalibration, demonstrate the need for multicalibration to achieve unbiased estimates of prevalence in out-of-domain settings, and apply those insights to a few real-world datasets.

Results

Standard calibration methods fail under covariate shift

Consider a binary outcome $Y \in \{0, 1\}$, features X , and a device $h(X) \in [0, 1]$ producing probabilistic predictions. The goal is to estimate the population prevalence $\pi = P(Y = 1)$ in a target population using only unlabeled target data $\{X_i\}$ and the device h . The key assumption concerns which aspect of the data-generating process changes across populations. Three types of distributional change are commonly distinguished:

- **Covariate shift:** The marginal distribution of features $P(X)$ changes, but the conditional $P(Y|X)$ remains stable. This is natural when features causally influence outcomes ($X \rightarrow Y$): applying a device to a new population changes the mix of features but not the underlying feature-outcome

¹<https://openai.com/index/using-gpt-4-for-content-moderation/>

Needed? Distractive to the reader? Compress?
 Perhaps focus only
 on explaining covariance
 shift, as this is only what is being used,
 and mention label shift and concept shift in passing.

relationships. Many measurement applications fit this pattern—demographics influence employment, text content determines topic relevance, patient characteristics drive diagnoses.

- **Label shift** (prior probability shift): The prevalence $P(Y)$ changes, and the conditional $P(X|Y)$ remains stable. This is natural when outcomes generate features ($Y \rightarrow X$), as in some diagnostic settings where a disease produces symptoms. However, for most measurement tasks where the goal is to classify based on observed features, the $X \rightarrow Y$ direction is more appropriate.
- **Concept drift**: The conditional $P(Y|X)$ itself changes—for example, the meaning of words evolves over time, or a disease mutates to present differently. No purely statistical post-processing can correct for concept drift; new labeled data from the target distribution is required.

We focus on covariate shift, which is the dominant concern in the measurement settings motivating this paper (see Limitations for further discussion of concept drift).

A device is *globally calibrated* if $\mathbb{E}[Y | h(X) = p] = p$ for all prediction values p . Under global calibration, $\mathbb{E}[h(X)] = \mathbb{E}[Y] = \pi$, so the sample mean of predictions is an unbiased prevalence estimate. However, global calibration is a property of a *specific* population. A device calibrated on one population need not be calibrated on another, even under covariate shift with stable $P(Y|X)$.

The failure mechanism is straightforward. Suppose the population consists of subgroups indexed by g , with sizes w_g and group-specific calibration errors $\epsilon_g = \mathbb{E}[h(X) - Y | G = g]$. Global calibration requires only that these errors cancel on average: $\sum_g w_g \epsilon_g = 0$. Under covariate shift, the weights change to w_g^* , and the bias in the prevalence estimate becomes $\sum_g w_g^* \epsilon_g$, which is generally nonzero unless every $\epsilon_g = 0$. Crucially, a device can be perfectly globally calibrated—with errors that precisely cancel in the training population—while being substantially miscalibrated on every individual subgroup.

The quantification literature has proposed several methods to correct for device errors (González et al. 2017). Each is vulnerable to the failure mechanism described above. *Classify & Count* binarizes predictions at a threshold τ and reports the positive fraction; the threshold is chosen to match true prevalence on calibration data, but this correspondence breaks under shift. The *Adjusted Count* (Rogan-Gladen) estimator $\hat{\pi} = (\hat{p} - \text{FPR}) / (\text{TPR} - \text{FPR})$ uses sensitivity and specificity from calibration data (Rogan and Gladen 1978). These error rates are weighted averages of group-specific rates; when group proportions shift, the learned rates no longer match the target population’s. The ratio estimator further amplifies errors when $\text{TPR} - \text{FPR}$ is small. *Probabilistic Adjusted Classify & Count* (PACC) generalizes this to soft scores, using $\mathbb{E}[h(X)|Y=1]$ and $\mathbb{E}[h(X)|Y=0]$ in place of binary error rates, but fails for the same reason: these conditional means are weighted averages over subgroups and shift with population composition.

The *Saerens-Latinne-Decaestecker* (SLD) algorithm (Saerens, Latinne, and Decaestecker 2002)—also known as EMQ—uses expectation-maximization to iteratively re-estimate prevalence by adjusting posteriors for a new prior; it is the strongest standard quantification method under label shift (where $P(X|Y)$ is stable), but under covariate shift, the class-conditional score distributions $P(h(X)|Y)$ change, causing SLD to converge to a biased estimate. The same argument applies to calibration approaches that assume stable error properties, such as the Calibrate-Extrapolate framework (Wu and Resnick 2024). Indeed, Wu and Resnick (2024) explicitly demonstrate that “covariate shift does not imply a stable calibration curve”: the calibrated posterior $P(Y=1 | h(X)=v)$ depends on the composition of individuals mapped to each score value v , which changes when $P(X)$ shifts. Their recommended estimator for the covariate-shift setting—averaging calibrated posteriors via isotonic regression—is precisely the global calibration baseline we evaluate below; as our results show, it still produces substantial bias because global calibration does not guarantee feature-conditional accuracy.

Multicalibration guarantees unbiased prevalence estimation

The analysis above shows that all standard methods produce biased prevalence estimates under covariate shift, with bias $\sum_g w_g^* \epsilon_g$ that is generally nonzero unless every $\epsilon_g = 0$. The illustration used discrete groups

for clarity, but the insight generalizes: what matters is that the device’s predictions are correct *conditional on the features* X , not just on average.

Multicalibration (Hébert-Johnson et al. 2018) formalizes this requirement. In its original formulation, a predictor $f(X)$ is α -multicalibrated with respect to a collection of subgroups $\mathcal{G}$ if $|\mathbb{E}[Y \mid f(X) = v, X \in G] - v| \leq \alpha$ for every $G \in \mathcal{G}$ and prediction value v . When $\mathcal{G}$ is rich enough to capture all relevant structure in X , this is equivalent to requiring calibration conditional on the full feature space:

$$\mathbb{E}[Y \mid f(X) = v, X = x] = v$$

for all x and v with sufficient probability mass. This is the framing of Kim et al. (2022)’s “Universal Adaptability”: a predictor calibrated conditional on X yields correct expected values under any reweighting of $P(X)$, without requiring knowledge of the shift.

The connection to prevalence estimation is direct. If f is calibrated conditional on X , then $\mathbb{E}[f(X) \mid X = x] = \mathbb{E}[Y \mid X = x]$ for all x . Under covariate shift—where only $P(X)$ changes while $P(Y|X)$ remains stable—the law of iterated expectations gives:

Def?
$$\underline{\mathbb{E}}^*[f(X)] = \int \mathbb{E}[f(X) \mid X = x] \underline{dP}^*(x) = \int \mathbb{E}[Y \mid X = x] \underline{dP}^*(x) = \underline{\mathbb{E}}^*[Y] = \underline{\pi}^*.$$

Because predictions are correct at each point in the feature space, the average of predictions tracks the true prevalence under *any* change in $P(X)$. Returning to the discrete illustration from the previous section: global calibration only ensures $\sum_g w_g \epsilon_g = 0$ for the training weights w_g ; calibration conditional on X ensures $\epsilon_g \approx 0$ everywhere, making the estimate robust to arbitrary changes in the feature distribution. This is also a *necessary* condition: if predictions are miscalibrated for any region of the feature space, there exists a target distribution that overweights that region and produces biased prevalence estimates.

Practical post-hoc multicalibration algorithms can be applied to any base predictor—including LLMs and black-box classifiers—by iteratively correcting predictions in regions of the feature space where they are miscalibrated (Hébert-Johnson et al. 2018; Gopalan et al. 2022). Early algorithms required discrete groups to be specified explicitly, but more recent methods take a set of features as input and automatically discover miscalibrated regions, effectively approximating calibration conditional on the full feature space with increasing data (Tax et al. 2025)².

Simulation: standard methods fail, multicalibration succeeds

We illustrate the theoretical results above with a controlled simulation. We generate data with a binary covariate $X \in \{0, 1\}$ and binary outcome Y , where $P(Y = 1|X = 1) = 0.85$ and $P(Y = 1|X = 0) = 0.15$. A classifier produces systematically biased predictions: 10% underestimation when $X = 0$ and 10% overestimation when $X = 1$. All calibration parameters are learned on a balanced training distribution ($P(X = 0) = 0.5$). Prevalence is estimated by averaging the predicted probabilities over the target sample. We then evaluate prevalence estimates on test distributions where $P(X = 0)$ ranges from 0.01 to 0.99, repeating 50 times (details in Materials and Methods). Def \Delta P(X = 0) here

Def? Figure 1: Prevalence estimation under covariate shift for six methods across 50 simulation runs. (a) Average bias (%). (b) Root mean squared error (RMSE). Only multicalibration maintains near-zero bias and RMSE across all levels of distribution shift.

Figure 1 shows the results. At the training distribution (center), all methods except the uncalibrated baseline produce approximately unbiased estimates. As the distribution shifts, the methods diverge.

²An open-source implementation is available at <https://mcgrad.dev>.

Prevalence Estimation Under Distribution Shift: Bias and RMSE

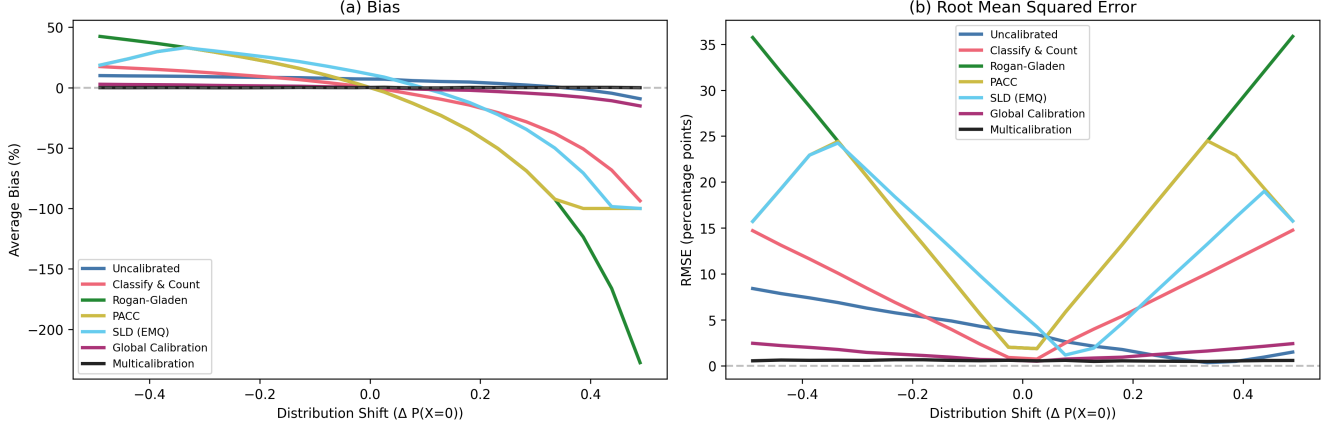


Figure 1: Figure 1

Classify & Count and Rogan-Gladen show rapidly growing bias, exceeding $\pm 40\%$ at extreme shifts. The Rogan-Gladen estimator is particularly unstable: its ratio structure amplifies estimation errors, producing bias that exceeds the other methods by an order of magnitude. SLD tracks the Rogan-Gladen curve closely—its EM procedure converges to a similarly biased estimate because the class-conditional score distributions, which SLD assumes are stable, shift under covariate shift.³ Global calibration shows more moderate but still substantial bias (up to $\pm 15\%$). The RMSE panel (Figure 1b) shows that these bias differences dominate total estimation error: the RMSE curves closely track the absolute bias, confirming that variance is small relative to bias for all methods at this sample size. The multicalibrated estimator maintains near-zero bias and the lowest RMSE across the entire range of distribution shifts.

Empirical application: employment prevalence under age distribution shift

To illustrate how distribution shift can lead to strongly biased measurements, we analyze data from the American Community Survey (ACS), a large-scale annual survey conducted by the U.S. Census Bureau. The concept to be measured is the rate of employment. The measurement device is a predictive model of binary employment status, with 16 sociodemographic features including age, education, marital status, disability status, and citizenship.

Setup. We train a logistic regression classifier on data from eight U.S. states (TX, MI, PA, OH, IL, GA, NC, VA) across 2016–2018, totaling approximately 1.5 million training observations. The remaining in-distribution data is split into a calibration set ($n \approx 644,000$) and a test set ($n \approx 920,000$). Six additional states (CA, NY, FL, WA, AZ, CO) are held out for out-of-distribution (OOD) evaluation. We fit two post-hoc calibration methods on the calibration set: isotonic regression (global calibration) and MCGrad, a multicalibration algorithm that enforces calibration conditional on both categorical and numerical features. We compare seven prevalence estimation methods: raw (uncalibrated) scores, Classify & Count with a prevalence-matched threshold, Rogan-Gladen adjustment, PACC, SLD, isotonic regression, and MCGrad.

Age distribution shift. Employment rates vary dramatically by age: approximately 47% for ages 16–24, 76% for ages 25–54, 61% for ages 55–64, and 17% for ages 65+. This makes age an ideal dimension along which to construct meaningful distribution shifts. We create synthetic target populations by resampling test data with shifted age distributions: young-skewed (oversampling ages 16–30), old-skewed (oversampling

³SLD’s bias is non-monotonic in the magnitude of shift: it grows at moderate shifts but decreases slightly at the extremes. This occurs because SLD’s output is bounded in $[0, 1]$ (as a mean of adjusted posteriors), so it saturates at 0 or 1 under extreme shift, while the true prevalence asymptotes to 0.15 or 0.85. The gap between these limits is smaller than the gap at intermediate shifts where SLD has not yet saturated.

ages 60+), and bimodal (oversampling both tails). The resulting populations have true employment rates ranging from 12.8% to 46.0%. All calibration parameters are estimated once on the original calibration set and held fixed across scenarios.

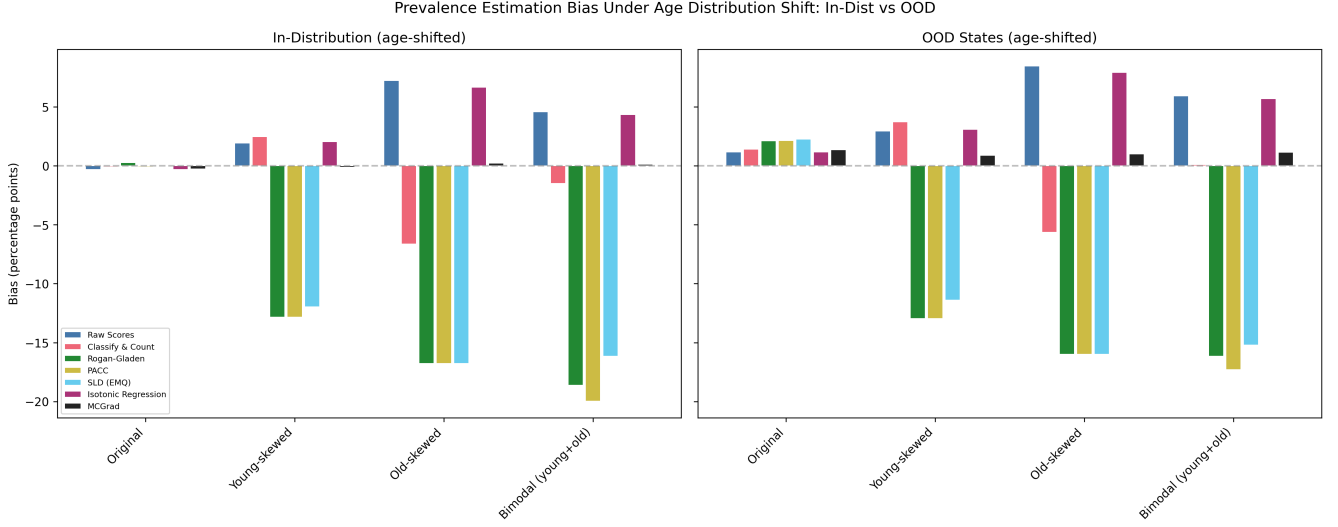


Figure 2: Figure 2

Figure 2: Prevalence estimation bias (percentage points) under synthetic age distribution shift, for in-distribution data (left) and out-of-distribution states (right).

Setting	Age Dist.	True Prev.	Raw	CC	RG	PACC	SLD	Iso.	MCGrad
In-Dist	Original	46.0%	-0.31	-0.07	+0.26	-0.07	+0.01	-0.30	-0.27
In-Dist	Young-skewed	12.8%	+1.93	+2.47	-12.82	-12.82	-11.95	+2.04	-0.11
In-Dist	Old-skewed	16.8%	+7.23	-6.62	-16.77	-16.77	-16.76	+6.65	+0.22
In-Dist	Bimodal	21.1%	+4.57	-1.50	-18.62	-19.97	-16.14	+4.33	+0.12
OOD	Original	45.1%	+1.15	+1.40	+2.12	+2.13	+2.25	+1.17	+1.35
OOD	Young-skewed	13.0%	+2.93	+3.73	-12.96	-12.96	-11.38	+3.08	+0.88
OOD	Old-skewed	16.0%	+8.47	-5.64	-15.97	-15.97	-15.97	+7.91	+1.01
OOD	Bimodal	20.8%	+5.91	+0.09	-16.14	-17.27	-15.20	+5.69	+1.13

Table 1: Prevalence estimation bias in percentage points (pp) under synthetic age distribution shift. RG = Rogan-Gladen, Iso. = Isotonic regression.

Results. With no age shift, all methods produce approximately unbiased estimates (Table 1). Under shift, the methods diverge sharply.

Rogan-Gladen exhibits catastrophic failure, with bias of -12.8 to -18.6 percentage points in the in-distribution setting—effectively estimating zero or negative prevalence when the true rate is 13–21%. PACC and SLD show comparable bias (-12 to -20pp and -11 to -16pp respectively), confirming the theoretical prediction: all three methods rely on calibration quantities—binary error rates, conditional score means, or class-conditional score distributions—that shift with population composition. Isotonic regression and raw scores show more moderate but still substantial bias of 2–8 percentage points, reflecting the instability of global calibration curves under compositional change. Classify & Count shows bias of up to 6.6 percentage points as the learned threshold loses its validity.

MCGrad produces near-zero bias across all in-distribution age-shift scenarios ($\leq 0.27\text{pp}$). Because MCGrad is multicalibrated with respect to age and other features, its predictions are correct *conditional on* these covariates. When the age distribution shifts, the feature-conditional predictions remain valid, and their average tracks the true prevalence. Bootstrap resampling (200 iterations per scenario) confirms that MCGrad also achieves the lowest RMSE, indicating that the bias reduction does not come at the cost of increased variance.

In the OOD setting—applying the model to states never seen during training *and* shifting the age distribution—MCGrad’s advantage persists with modestly larger bias (0.88–1.35pp), reflecting the additional geographic shift along a dimension not included in the calibration features. Even so, MCGrad’s bias remains 3–8× smaller than the next best method.

LLM-based topic classification under cross-national shift

The ACS application uses a standard ML classifier with well-distributed probability estimates. We now examine a more challenging and increasingly common setting: using an LLM as a zero-shot measurement device, where the score distribution is highly non-standard and the target populations span multiple countries and languages.

Setup. We use Llama 3.3 70B Instruct (Dubey et al. 2024) (4-bit quantized) to classify political texts from the Comparative Agendas Project (Baumgartner, Green-Pedersen, and Jones 2006) as related to Law & Crime (CAP major topic code 12). The data comprise 105,000 documents from six sub-populations across four countries and four languages: Danish parliamentary questions, Spanish oral questions, U.S. congressional bills, Belgian newspaper articles, Spanish media articles (El Pais and El Mundo), and Belgian TV news. LLM scores are extracted via token log-probabilities: $h(X) = P(\text{Yes}) / (P(\text{Yes}) + P(\text{No}))$. The classifier achieves strong discriminative performance across most sub-populations (AUC 0.878–0.955), with the exception of Danish parliamentary questions (AUC 0.745).

The calibration set draws equally from four sub-populations (Denmark questions, Spain questions, U.S. bills, Belgium newspaper; 10,000 each, 40,000 total), ensuring that all four countries and all four languages are represented. MCGrad is calibrated with categorical features (country, document type, party) and one numerical feature (decade).⁴ Two sub-populations are held out as out-of-distribution targets: Spanish media (same country and language as calibration, but unseen document type) and Belgian TV news (same country and language as calibration, but unseen document type). This design mirrors a realistic deployment scenario—validating a measurement device on one document type and applying it to another within the same country—and isolates document type as the single novel dimension of shift.

Score characteristics. Unlike the ACS classifier, which produces well-distributed probability estimates, the LLM produces highly bimodal scores: 79–94% of positive documents score above 0.99 (47% for Denmark, where the LLM is less confident), while 53–78% of negatives score below 0.01 (Figure 3). Critically, 10–24% of negative documents also score above 0.99—the LLM is confidently wrong on a substantial fraction of negatives, particularly for U.S. bills (24%). This extreme distribution requires preprocessing before multicalibration: MCGrad’s internal logit transform maps values near 0 and 1 to $\pm\infty$, causing numerical instability. We apply a linear squashing transformation $h'(X) = \epsilon + (1 - 2\epsilon) \cdot h(X)$ with $\epsilon = 0.05$ to compress scores into $[0.05, 0.95]$ before fitting MCGrad. This preserves rank ordering without introducing calibration assumptions—MCGrad itself performs the calibration. Squashing is applied only to the MCGrad pipeline; SLD and all other methods operate on raw scores, since SLD interprets its input

⁴We also tested a richer feature set by adding 500 PCA components of the LLM’s last-layer hidden states (8192-dim) as additional numerical segment features. The embedding features produced negligible changes in bias and RMSE across all scenarios, suggesting that the information they encode is largely redundant with the LLM’s classification score. See SI Appendix for details.

as posteriors and squashing distorts this interpretation.⁵ The bimodal score distribution independently causes SLD’s EM algorithm to converge to a biased fixed point, as the scores violate its assumption of calibrated posteriors.

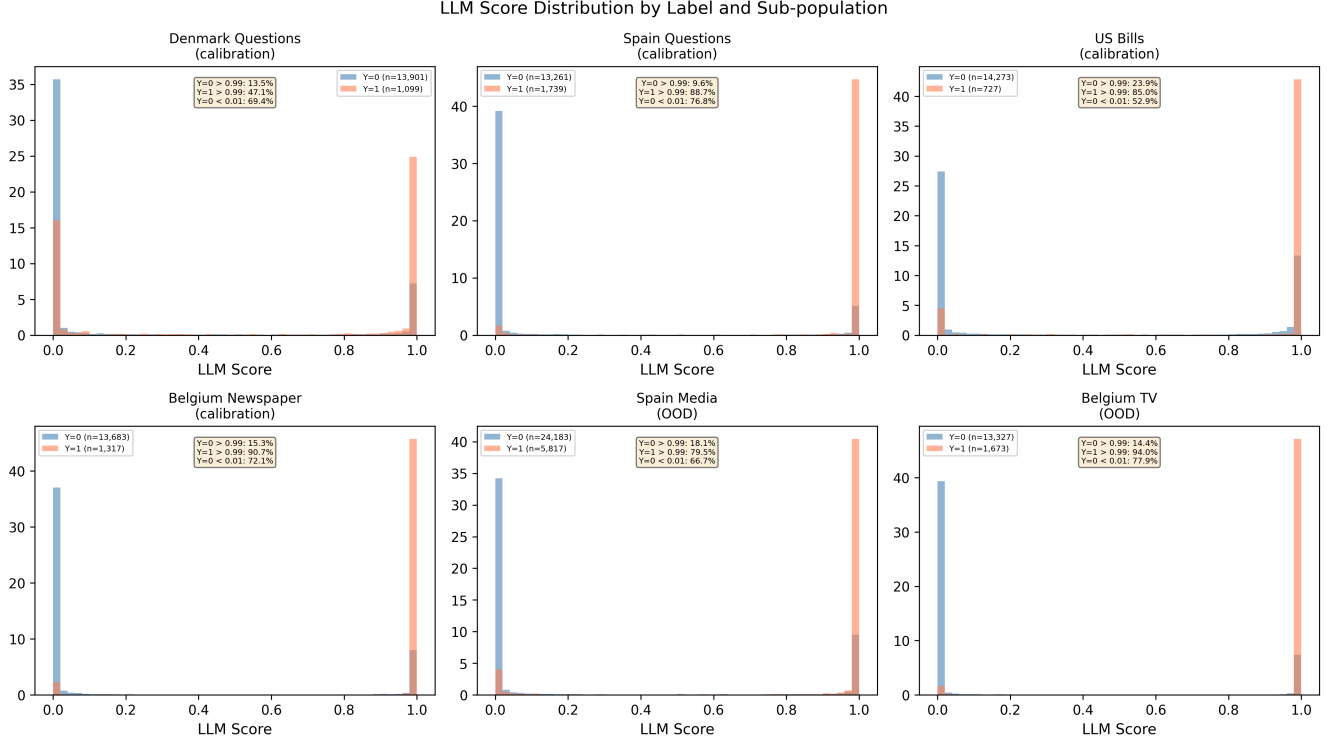


Figure 3: Figure 3

Figure 3: LLM score distribution by label across six CAP sub-populations. Scores are highly bimodal, with substantial false-confident negatives (scores > 0.99 for $Y=0$), especially in U.S. bills (24%).

Results. The shift gradient spans six scenarios (Table 2, Figure 4): a baseline with no shift, three within-calibration shifts (country composition, document type composition, party composition), and two OOD scenarios where document type is the single novel dimension.

Scenario	Shift Type	True Prev.	Raw	CC	RG	PACC	SLD	Iso.	MCGRad
Baseline	None	8.1%	+19.9	+11.5	-0.2	-0.3	+22.7	-0.0	-0.1
Country shift	Within-cal.	8.7%	+19.1	+12.6	+1.9	-1.2	+21.6	+0.5	-0.2
Doc-type shift	Within-cal.	6.5%	+27.7	+16.5	+6.8	+11.4	+31.4	+2.6	+0.1
Party shift	Within-cal.	9.0%	+19.0	+11.1	-0.2	-1.3	+21.7	-0.9	+0.0

⁵We verified that applying squashing to SLD’s input makes its performance substantially worse: because squashing compresses scores toward the center of $[0,1]$, SLD’s EM algorithm interprets the inflated posteriors as evidence of much higher prevalence, producing bias exceeding +70pp in most scenarios.

Scenario	Shift Type	True Prev.	Raw	CC	RG	PACC	SLD	Iso.	MCGrad
Spain media	OOD doc type	19.3%	+17.5	+10.0	+3.9	+2.8	+20.9	-8.0	-7.8
Belgium TV	OOD doc type	11.1%	+15.1	+11.6	+1.7	-6.2	+16.3	-0.4	-1.6

Table 2: Prevalence estimation bias (pp) for Law & Crime topic under cross-national and document-type shift. Bootstrap RMSE (200 iterations) closely tracks absolute bias in all scenarios. RG = Rogan-Gladen, Iso. = Isotonic regression.

Figure 4: Prevalence estimation bias (pp) across the shift gradient. MCGrad achieves near-zero bias within the calibration distribution. On OOD populations (unseen document type), MCGrad’s bias increases modestly for Belgian TV but more substantially for Spanish media.

The most striking result is the failure of raw LLM scores. Despite strong discriminative performance (AUC 0.878–0.955 across most sub-populations), simply averaging the LLM’s predicted probabilities produces prevalence estimates biased by +15 to +28 percentage points—even at baseline, with no distributional shift at all. The true Law & Crime prevalence is 8.1%, but the raw scores estimate approximately 28%. This disconnect between discrimination and calibration has immediate practical implications: researchers who validate an LLM classifier by reporting high AUC or accuracy, then estimate prevalence by averaging its scores over a corpus, will obtain grossly inflated estimates regardless of how well the model separates positive from negative documents. The bias arises because the LLM’s bimodal score distribution is systematically shifted upward—a property invisible to rank-based evaluation metrics. SLD performs even worse than raw scores because its EM algorithm converges to a biased fixed point when the bimodal LLM scores violate its calibrated-posterior assumption—a failure mode distinct from the covariate-shift vulnerability identified in the ACS application.

Within the calibration distribution, MCGrad achieves near-zero bias across all scenarios: ≤ 0.2 pp under country shift, document-type shift, and party shift. This represents a clear improvement over isotonic regression, which shows bias of 0.5–2.6pp under the same shifts—consistent with the theoretical prediction that multicalibration provides stronger guarantees than global calibration when the shift is along calibrated feature dimensions. The contrast is sharpest under document-type shift, where overweighting U.S. bills changes the score distribution substantially: MCGrad maintains +0.1pp bias while isotonic degrades to +2.6pp.

On OOD populations—where document type is the single novel dimension—the results depend on the severity of the shift. For Belgian TV news (true prevalence 11.1%), MCGrad estimates 9.6%, a bias of only -1.6pp. This is a realistic transfer scenario: the model was validated on Belgian newspaper articles and applied to Belgian TV news from the same country and language. MCGrad’s country-level corrections, learned from Belgian newspaper data in calibration, transfer reasonably well to a related document type. Isotonic regression performs slightly better here (-0.4pp), but both methods produce usable estimates.

For Spanish media (true prevalence 19.3%), both calibration methods show larger bias: MCGrad at -7.8pp and isotonic at -8.0pp. This scenario is harder because it combines an unseen document type with a substantial prevalence difference—the true prevalence in Spanish media (19.3%) is more than double the calibration prevalence (8.1%). MCGrad’s country-level corrections, learned from Spanish parliamentary questions (11.6% prevalence), underestimate the higher prevalence of Law & Crime in Spanish media coverage. Rogan-Gladen (+3.9pp) and PACC (+2.8pp) perform better here because their calibration-set

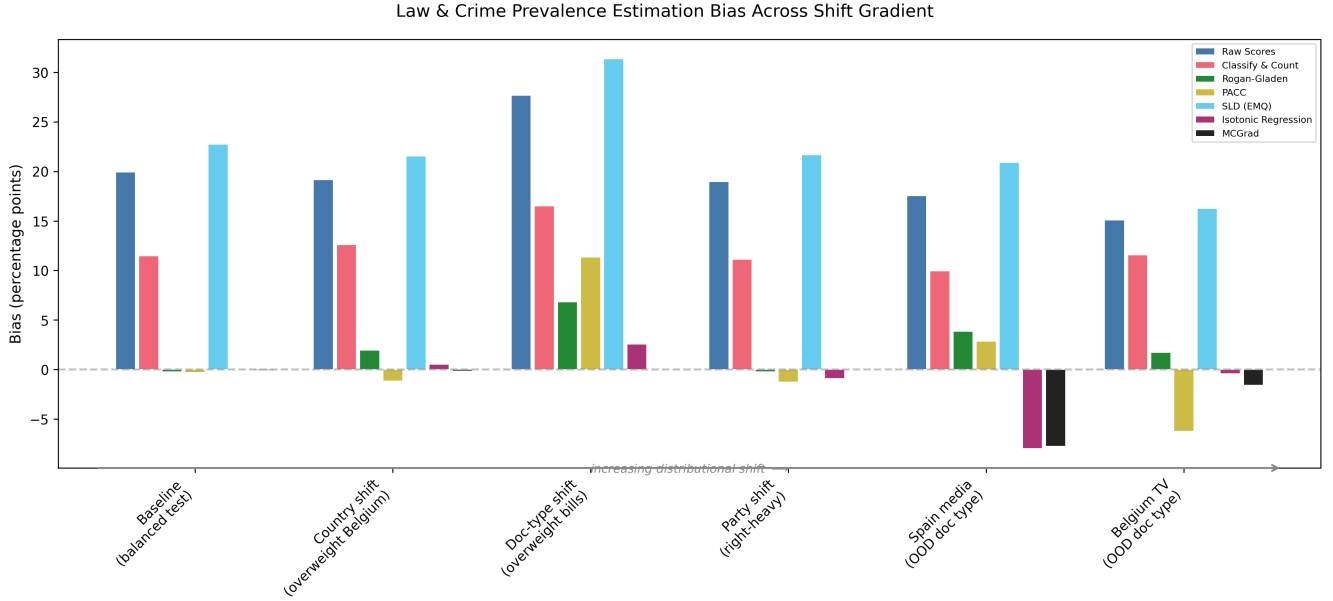


Figure 4: Figure 4

error rates coincidentally approximate the target-population rates for this specific scenario—not because of any systematic robustness, as their within-calibration failures confirm.

Comparison with the ACS application. Across both applications, the pattern is consistent: MCGrad provides the strongest bias reduction when the target population’s features are represented in the calibration set, and degrades gracefully when one dimension is novel (ACS OOD states: $\leq 1.35\text{pp}$; CAP Belgian TV: -1.6pp). The larger bias on Spanish media (-7.8pp) reflects the combination of an unseen document type with a large prevalence difference, exceeding what any single method handles well.

Discussion

We have shown that multicalibration—calibration conditional on the input features, not just on average—is the key property for accurate model-based prevalence estimation under population shift. The theoretical guarantee is precise: multicalibration is necessary and sufficient for unbiased estimation under covariate shift. In practice, this requires that the target population’s features lie within the support of the calibration distribution. Both empirical applications confirm this: when calibrated features overlap with the target population, MCGrad achieves near-zero bias; when the target introduces a novel feature value, bias increases with the severity of the shift. Practitioners should ensure their calibration data covers the key dimensions along which target populations may differ.

The practical implications are immediate. The growing use of LLMs as zero-shot measurement devices rests on the assumption that models validated on one population can be reliably applied to others. Our analysis shows this assumption is safe only when the model is multicalibrated with respect to features that may shift across populations. Without this, prevalence estimates in new populations—the primary use case for LLM-based measurement—are unreliable. Researchers using AI-based measurement should treat multicalibration not as an optional fairness criterion, but as a methodological requirement for valid inference.

This result also clarifies why certain widely-used practices are inadequate. The common approach of reporting a model’s overall accuracy, F1, or AUC on a held-out set provides no information about calibration, let alone multicalibration. A model with 95% accuracy can still produce heavily biased

prevalence estimates if its errors are unevenly distributed across the feature space. Similarly, reporting a single calibration curve—as is standard in the quantification literature (González et al. 2017)—certifies only global calibration, which we have shown is insufficient under shift. Even comprehensive methodology textbooks for text-as-data research devote extensive attention to classification accuracy and validation but do not discuss calibration as a requirement for valid prevalence estimates (Grimmer, Roberts, and Stewart 2022); multicalibration is entirely absent from these discussions.

Applying multicalibration to LLM-based measurement. Multicalibration does not require access to a model’s internals. When LLMs classify documents, survey responses, or clinical notes, observable metadata—document source, author demographics, text length, language—can serve directly as segment features, and these often correspond to the dimensions along which population composition may shift. For richer feature structure, Detommaso et al. (2024) demonstrate two approaches for constructing features over prompt-completion pairs: clustering within the embedding space and “self-annotation” via yes/no questions about input properties. Both yield features enabling effective multicalibration of LLM confidence scores, though end-to-end pipelines for LLM-based prevalence estimation remain an active area of development.

Several limitations warrant discussion. First, multicalibration requires specifying the features over which to calibrate; the guarantee holds only for shifts along included dimensions. As the CAP application demonstrates, performance degrades when target populations introduce novel feature values, particularly when compounded by a large prevalence difference. Practitioners should ensure calibration data covers the range of feature values likely to appear in target populations. Second, multicalibration requires labeled calibration data of sufficient size. This is feasible for standard ML pipelines but more challenging for zero-shot LLM settings, where calibration labels may require the manual annotation LLMs were intended to avoid; prediction-powered inference (Angelopoulos et al. 2023) offers a complementary framework. Third, LLMs produce text rather than probability scores; calibratable scores can be obtained via token log-probabilities, confidence-elicitation prompts, or repeated sampling (Detommaso et al. 2024). Fourth, our framework assumes covariate shift ($P(Y|X)$ stable); under concept drift, no purely statistical correction substitutes for new labeled data. Finally, both applications use settings where ground-truth labels enable direct bias measurement—synthetic shifts in the ACS case and expert-coded labels across countries in the CAP case.

These results connect two literatures that have developed in isolation. The quantification literature has focused on correcting aggregate error rates but has not engaged with feature-conditional calibration (González et al. 2017; Wu and Resnick 2024). The multicalibration literature has focused on individual-level prediction quality and fairness but has not emphasized the implications for population-level inference (Hébert-Johnson et al. 2018; Kim et al. 2022). Our contribution is to show that calibration conditional on the feature space is the missing link that makes model-based prevalence estimation reliable under the distribution shifts that motivate its use.

Materials and Methods

Simulation. We generate $n = 10,000$ observations with a binary covariate $X \in \{0, 1\}$ and outcome $Y \sim \text{Bernoulli}(P(Y|X))$, where $P(Y = 1|X = 1) = 0.85$ and $P(Y = 1|X = 0) = 0.15$. The classifier produces deterministic scores $\hat{p}(X = 0) = 0.135$ and $\hat{p}(X = 1) = 0.935$, representing 10% multiplicative bias within each stratum. For each of $B = 50$ iterations, we generate fresh calibration data from $P(X = 0) = 0.5$, estimate all method-specific parameters, then evaluate bias and mean squared error on 20 test distributions with $P(X = 0)$ ranging from 0.01 to 0.99. The seven methods compared—uncalibrated averaging, Classify & Count with a prevalence-matched threshold, Rogan-Gladen adjustment, PACC, SLD (EMQ), global multiplicative calibration, and multicalibration with stratum-specific additive corrections—are detailed with full mathematical definitions in the SI Appendix.

Empirical application (ACS). We use the ACS Public Use Microdata Sample via the *folktables* package, with the ACSEmployment prediction task (binary: employed vs. not employed) and 16 sociodemographic features. Training data comprises eight states (TX, MI, PA, OH, IL, GA, NC, VA) across 2016–2018. The base model is logistic regression with standard scaling. Post-hoc calibration uses isotonic regression (global) and MCGrad (multicalibration with categorical and numerical segment features) on a held-out calibration set ($n \approx 644,000$). In addition to raw scores, Classify & Count, Rogan-Gladen, isotonic regression, and MCGrad, we include two methods from the quantification literature: PACC (Probabilistic Adjusted Classify & Count), which generalizes Rogan-Gladen to continuous scores (González et al. 2017), and SLD (Saerens-Latinne-Decaestecker), an EM algorithm that iteratively adjusts posteriors for a new prior (Saerens, Latinne, and Decaestecker 2002). Synthetic age-shifted populations are created by importance-weighted resampling of the test set ($n \approx 920,000$), with exponential weights favoring young ages, old ages, or both extremes. RMSE is computed via bootstrap resampling (200 iterations per scenario). Six additional states (CA, NY, FL, WA, AZ, CO) serve as OOD evaluation data.

Empirical application (CAP). We use data from the Comparative Agendas Project (Baumgartner, Green-Pedersen, and Jones 2006), which provides expert-coded policy topic labels for political texts across countries. The binary outcome is whether a document addresses Law & Crime (CAP major topic code 12). We sample 105,000 documents: 15,000 from each of six sub-populations (Danish parliamentary questions, Spanish oral questions, U.S. congressional bills, Belgian TV news, Belgian newspaper articles) and 30,000 from Spanish media (El Pais and El Mundo combined). The measurement device is Llama 3.3 70B Instruct (Dubey et al. 2024), run with 4-bit NF4 quantization on a single A100 80GB GPU. For each document, we prompt the model with the CAP codebook definition of Law & Crime and extract a binary score via token log-probabilities: $h(X) = \exp(\log P(\text{Yes})) / [\exp(\log P(\text{Yes})) + \exp(\log P(\text{No}))]$. The LLM produces highly bimodal scores (69–94% of positives above 0.99, 53–78% of negatives below 0.01), which cause numerical instability in MCGrad’s internal logit transform. We apply a linear squashing transformation $h'(X) = \epsilon + (1 - 2\epsilon) \cdot h(X)$ with $\epsilon = 0.05$ to compress scores into $[0.05, 0.95]$ before fitting MCGrad; this preserves rank ordering without introducing calibration assumptions. Squashing is applied only to the MCGrad pipeline; all other methods operate on raw scores. Results are robust to the choice of ϵ (tested at 0.01, 0.05, and 0.10; see SI Appendix). The calibration set ($n \approx 40,000$) draws equally from four sub-populations—Denmark questions, Spain questions, U.S. bills, and Belgium newspaper—ensuring that all four countries and all four languages are represented. MCGrad is calibrated on squashed scores with categorical features (country, document type, political party) and one numerical feature (decade). Isotonic regression is fitted on raw scores using the same calibration set. The threshold for Classify & Count and Rogan-Gladen is set via Youden’s J statistic (maximizing TPR – FPR on the calibration set). PACC parameters ($\mathbb{E}[h(X)|Y=1]$ and $\mathbb{E}[h(X)|Y=0]$) and SLD’s source prevalence are estimated on the calibration set. Bootstrap RMSE uses 200 iterations per scenario. Two sub-populations are held out as out-of-distribution targets: Spanish media and Belgian TV news, which differ from the calibration data only in document type (media articles and TV news vs. parliamentary questions, bills, and newspaper articles), while sharing the same countries and languages seen during calibration.

Software. MCGrad is available at github.com/facebookincubator/MCGrad. Simulation and analysis code are available at [TODO: insert repository URL before submission].

Author Contributions

[TODO: Add author contributions, e.g., “A.B. designed research; C.D. performed research; E.F. contributed analytic tools; A.B. and C.D. analyzed data; and A.B. wrote the paper.”]

Competing Interests

The authors declare no competing interests.

Data Availability

The American Community Survey data are publicly available via the *folktables* package. The Comparative Agendas Project data are publicly available at <https://www.comparativeagendas.net>. Simulation and analysis code are available at [TODO: insert repository URL].

Keywords

multicalibration, prevalence estimation, covariate shift, large language models, quantification

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